

Powers of 10 & Standard Form

Standard (scientific) form is a tidy way to write very large or very small numbers using powers of 10 — so you never miscount zeros.

What is standard form?

The idea: Every number can be written as $x \times 10^y$, where **x is at least 1 and less than 10** (so exactly one non-zero digit before the decimal point) and **y is an integer**.

$$5900 = 5.9 \times 10^3 \quad | \quad 0.00037 = 3.7 \times 10^{-4}$$

How to write a number in standard form

1. Put the decimal point just after the **first non-zero digit** to get x.
2. Count how many places the point moved. That count is the power y.
3. **Large number** (point moved left) → **positive** power. **Small decimal** (point moved right) → **negative** power.

Expanded form with powers of 10

$$47561 = (4 \times 10^4) + (7 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$$

Comparing numbers in standard form

First compare the **power of 10**; the bigger power wins. If the powers are equal, compare the coefficients x.

Prerequisite refresher: $10^1 = 10$, $10^2 = 100$, $10^3 = 1000 \dots$; and $10^{-1} = 0.1$, $10^{-2} = 0.01$, $10^{-3} = 0.001 \dots$

Where marks slip away

Trap 1 — x out of range. In standard form x must be ≥ 1 and < 10 . So 59×10^2 is NOT standard form; 5.9×10^3 is.

Trap 2 — Wrong sign on the power. Big numbers get a positive power; small decimals get a negative power. $0.0004 = 4 \times 10^{-4}$, not 4×10^4 .

Trap 3 — Miscounting places. Count the decimal-point moves carefully — that count is the exponent.

Trap 4 — Comparing only the front digits. 9.35×10^6 is smaller than 6.27×10^7 , because $10^6 < 10^7$. Check the power first.

Slip 5 — Tidy the product. After multiplying, e.g. 13.12×10^5 , rewrite so $x < 10$: 1.312×10^6 .

Model answers — write them like this

Example A. Write 47561 in expanded form using powers of 10.

Step 1: Give each digit its place value as a power of 10.

Step 2: $47561 = (4 \times 10^4) + (7 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$.

$$\therefore 47561 = (4 \times 10^4) + (7 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$$

Example B. Write 39,00,00,000 in standard form.

Step 1: The number is 3900000000. Put the point after the first digit: 3.9.

Step 2: The point moved 8 places to the left $\rightarrow$ power = 8 (positive, large number).

$$\therefore 39,00,00,000 = 3.9 \times 10^8$$

Example C. Write 0.000005678 in standard form.

Step 1: First non-zero digit is 5; write 5.678.

Step 2: The point moved 6 places to the right $\rightarrow$ power = -6 (small number).

$$\therefore 0.000005678 = 5.678 \times 10^{-6}$$

Example D. Express $3.2 \times 10^6 \times 4.1 \times 10^{-1}$ in standard form.

Step 1: Multiply the coefficients: $3.2 \times 4.1 = 13.12$.

Step 2: Add the powers of 10: $10^6 \times 10^{-1} = 10^5$. So we have 13.12×10^5 .

Step 3: Tidy so $x < 10$: $13.12 \times 10^5 = 1.312 \times 10^6$.

$$\therefore = 1.312 \times 10^6$$

Example E. Which is larger: 1.4335×10^{12} or 1.439×10^{12} ?

Step 1: The powers are equal (both 10^{12}), so compare the coefficients.

Step 2: $1.439 > 1.4335$.

$\therefore 1.439 \times 10^{12}$ is larger

Next: [try the worksheet](#), [then check the answer key](#).